Alfredo Navarrete Montes

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EDUCATION

Benemerita Universidad Autonoma de Puebla (BUAP) Puebla, Mexico
Bachelor of Engineering in Information Technologies. *Expected Graduation: December 2026*
*Relevant Coursework (completed at **University of New Mexico, USA** - revalidated):* High Performance Computing, Information Systems Development, Computer-based Information Systems, Business Programming Fundamentals, Design of Large Programs.

EXPERIENCE

Volkswagen de Mexico (VWM) - IT Solutions Puebla, Mexico
Software Engineering Intern *August 2026 - Present*

- Developed a script with a two-flow architecture (ETL) that identifies CBD files by content, recognizing multiple template versions in English, Spanish and German, converts legacy .xls files to .xlsx and uploads them to AWS S3.
- Applied execution-time optimization strategies so the job handles large volumes of files in the background without interrupting the user.
- Implemented incremental state control with file fingerprints (modified date and size).
- Built the S3 upload stage using AWS CLI, working through the Zscaler corporate proxy, with rotating logs and failure-only alerts.
- Migrated authentication from a provisional technical user to IAM Roles Anywhere with a VW CA certificate and validated the upload in QAS and production. Coordinated with the S3 consumers and the AWS account administrator on naming, prefixes and permissions.

Health Technologies Laboratory - BUAP Puebla, Mexico
Research Assistant *January 2026 - August 2026*

- Participated in MentalRiskES 2026 (IberLEF 2026 shared task) on early detection of mental health risk from Spanish social media text, as part of the BUAP research team.
- Won 1st place in Task 1 (early detection category) and placed 6th in the Task 1 general category, with the best R10 score (MAE 0.889) and the lowest energy consumption among all participating teams.
- Built and ran the competition pipeline in Python, orchestrating multiple LLMs (GPT-4o-mini, GPT-4.1-nano, GPT-5-nano) through the OpenAI API with mock and dry-run workflows.
- Co-authored the resulting paper, accepted for publication in CEUR-WS (indexed in Scopus and DBLP).

University of New Mexico (UNM) - USA Albuquerque, NM
Research Assistant *August 2025 - December 2025*

- Designed a distributed Python ETL pipeline that collected 150k+ Kickstarter campaign records, enabling large-scale analysis of customer behavior.
- Developed automated scrapers in Python to extract structured and unstructured data from six JS-rendered sub-pages per campaign.
- Ran the crawl as independent workers over 250 rotating proxies, using static sharding (82 dataset shards) instead of a message broker, so any worker could be restarted without affecting the others.
- Treated Cloudflare challenges, CAPTCHAs and HTTP 429 limits as normal states (rotate proxy, rebuild browser profile, retry) and saved results after every row, so a crash lost at most one record.
- Built data preprocessing workflows to clean and merge datasets for analysis and machine learning experiments.
- Collaborated with graduate researchers at the Anderson School of Management to analyze customer behavior and market trends.

HIGH PERFORMANCE COMPUTING PROJECTS

HPC Benchmarking - Center for Advanced Research Computing (UNM) *August 2025 - December 2025*

- Benchmarked HPL on the Hopper cluster (32 cores per node, InfiniBand, GCC), tuning matrix size N, block size NB, and process grid P x Q through automated Slurm parameter sweeps.
- Raised throughput from 209 to 1,106 GFLOP/s at a fixed rank count by optimizing process grid shape, and reached 2.95 TFLOP/s on 128 MPI ranks.
- Ran CPU-only HPCG experiments on the Easley cluster, scaling from 1 to 128 cores across 4 nodes and analyzing serial fraction and parallel efficiency for a GPU vs CPU study.
- Conducted strong and weak scaling analysis (Amdahl and Gustafson) and co-authored two technical reports in ACM format.

TECHNICAL SKILLS

Programming: Python, C++, TypeScript, PowerShell, VBScript
Distributed Systems & HPC: Slurm, MPI, HPL, HPCG, Linux
Cloud & Automation: AWS S3, AWS CLI, IAM Roles Anywhere, Power Automate, SharePoint, Task Scheduler
Web Development: Angular, Django, REST APIs, MySQL
Data & Machine Learning: NLP, Transformers, LLM APIs, Data Preprocessing, Embeddings
Tools: Git, GitHub

LANGUAGES

Spanish: Native | **English:** Advanced (TOEFL ITP: 560, Duolingo English Test: 120)